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A paleomagnetic and AMS study of the Rätan granite of the Transscandinavian Igneous Belt, central Sweden

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Abstract: The 1.85–1.67 Ga Transscandinavian Igneous Belt (TIB) forms a major Paleoproterozoic igneous complex in the western part of the Fennoscandian Shield. Different tectonic models propose that the TIB was formed in a tectonic regime governed either by compression or by extension. This paper presents an analysis of anisotropy of magnetic susceptibility (AMS) and paleomagnetic data from the Rätan granite (1.70 Ga), which forms a large pluton in the central part of the TIB. The major aim of the study is to test the existing tectonic models and to define a paleomagnetic pole position for the Rätan granite. Three contrasting magnetic fabric domains are identified. In the eastern and western domains the AMS fabric is characterized by NW–SE to N–S trending subvertical foliation planes and a girdle distribution of the maximum and intermediate anisotropy axes. The central domain is characterized by subhorizontal magnetic foliation planes and lineations that cluster in a NW–SE trending direction. We suggest that the Rätan granite was emplaced in a tectonic regime governed by NW–SE directed extension and that the pluton was fed obliquely with a magma source situated to the present northwest. A positive baked contact test is demonstrated for a basic dyke suggesting that the Rätan granite has not been re-magnetized since c. 1.6 Ga. A paleomagnetic formation mean direction is defined for the granite (decl. = 2°, incl. = 59°) that yields a paleopole position of Plat = 67° and Plon = 190°.

Keywords: Anisotropy of magnetic susceptibility (AMS), paleomagnetism, Proterozoic magmatism, Transscandinavian Igneous Belt (TIB), Fennoscandian Shield.

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The Transscandinavian Igneous Belt (TIB) forms a major Late Paleoproterozoic igneous complex in the Fennoscandian Shield. The TIB was formed at c. 1.85–1.67 Ga (Ahl et al. 1999) along the southwestern boundary of the Svecofennian Domain (Gaál & Gorbatschev 1987). It defines a belt of igneous rocks that extends c. 700–800 km from the southern part of the Baltic Sea to the Caledonian front (Fig. 1) and continues northward up to the Lofoten islands in northern Norway (Åhäll & Larson 2000). It should be noted that some authors also include the Revsund granitoids in the TIB (e.g. Gorbatschev & Bogdanova 1993). The formation of TIB granitoids in the Svecofennian Domain suggests that these granitoids were formed in a destructive-margin process (e.g. Gorbatschev & Bogdanova 1993), and plate tectonic models proposed that the TIB was formed in a tectonic regime governed by compression (Windley 1993). However, Ahl et al. (1999) present a model suggesting that the Dala granitoids in the central part of the TIB were formed in a period of transition between subduction-related compression (1.81–1.77 Ga) and an extensive environment (1.70–1.68 Ga). The Rätan granite (e.g. Lundegårdh 1967; Lundqvist 1968), which forms a large intrusion north and east of the Dala granitoids, was emplaced at c. 1.70 Ga (U–Pb, Wilson et al. 1985; Patchett et al. 1987; Delin 1996). Sparse AMS data from the granite indicate that it may have formed in an extensive tectonic regime (Mattsson & Elming 2001).

In this study new AMS data from the Rätan granite are presented in order to better constrain the results from Mattsson & Elming (2001) and to test whether or not the granite was emplaced in an extensive tectonic regime. We also discuss new paleomagnetic results from the Rätan granitoid batholith. The southern and central parts of the TIB have been subject to fairly comprehensive geochronological, geochemical and also some AMS and paleomagnetic studies (e.g. Andersson 1991; Bylund & Elming 1992; Wikström et al. 1997; Lundqvist & Persson 1999; Åhäll & Larson 2000). However, in spite of the fact that the Rätan granite forms a large part of the TIB it has never been studied in detail. Plate tectonic reconstructions are important for a better understanding of the geological evolution of Fennoscandia and paleomagnetism is a well-known technique to perform such reconstructions. However, for the major part of the time of formation of the TIB the paleomagnetic data of the Fennoscandian Shield are very sparse (Elming et al. 1993). New paleomagnetic data from this time interval are therefore important in order to increase our knowledge of the plate tectonic history of the shield and the formation of the TIB.

General geology

The Fennoscandian Shield is commonly subdivided into three major domains (Gaál & Gorbatschev 1987); the Archean Domain in the northeast, the Svecofennian Domain in the central part of the shield, and the Southwest Scandinavian Domain in the southwest (Fig. 1, inset). The Svecofennian Domain was formed in the Early Proterozoic and accreted onto the Archean Domain during the Svecofennian orogeny at 1.93–1.87 Ga (e.g. Mellqvist et al. 1999). The Early Svecofennian granitoids were formed at c. 1.90–1.87 Ga and comprise I-type rocks of various chemical composition ranging from gabbros to granites (Gaál & Gorbatschev 1987). After this igneous activity a second generation of granitoid rocks, the Late Svecofennian granites, was formed. These rocks are related to the Svecofennian regional deformation and metamorphism (Ahl et al. 1999).

The TIB defines a north–south trending belt of igneous rocks, which form a complex of post-Svecofennian granitoids located along the boundary between the South Svecofennian Subprovince and the Southwest Scandinavian Domain. The rocks of the TIB are commonly subdivided into three separate groups (Larson & Berglund 1992): the Småland-Värmland granitoids,

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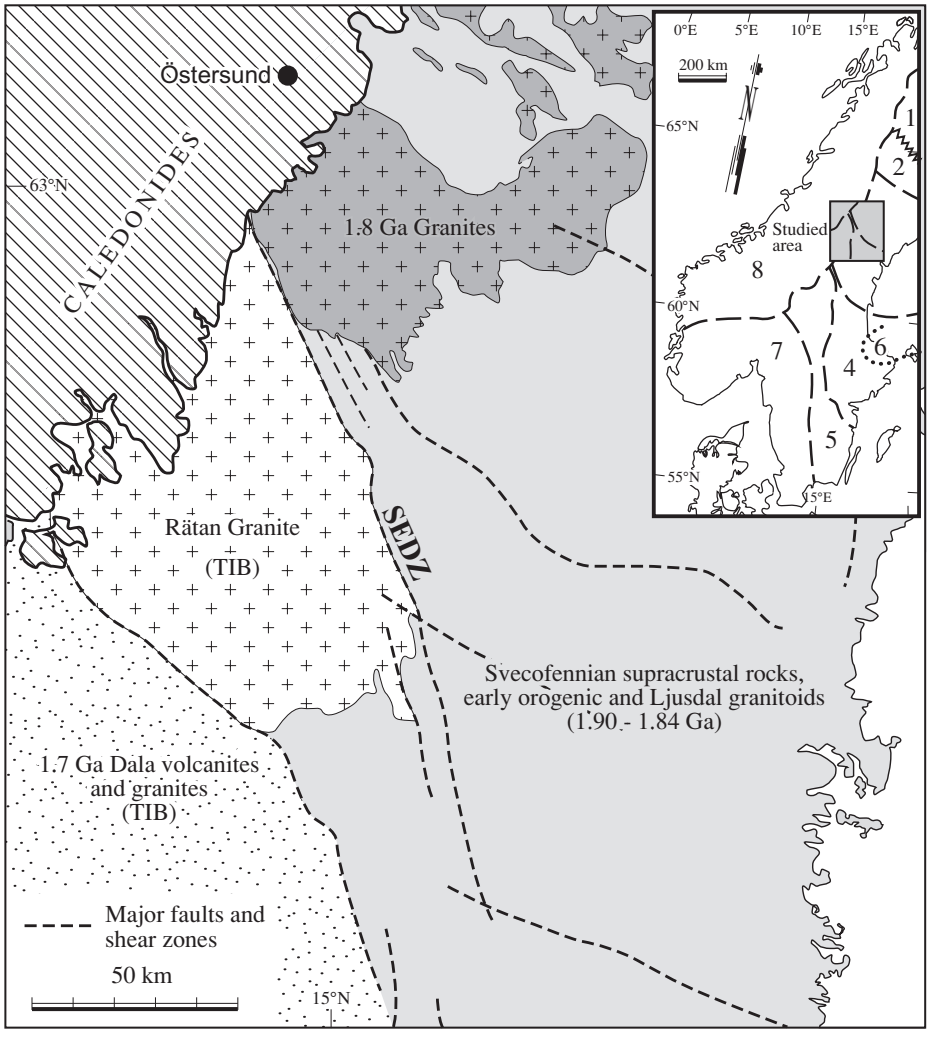


Fig. 1. Generalized geological map showing the tectonic setting and major deformation zones of the central parts of the Fennoscandian Shield, Sweden (re-drawn from Fredén 1994 and inferred from Bergman & Sjöström 1994 and Högdahl & Sjöström 1999). The SEDZ is the Storsjön-Edsbyn Deformation Zone. The inset figure shows the location of the study area. It also shows a tectonomagmatic block division of Fennoscandia (modified after Elming et al. 1993). 1, Northern Svecofennian subprovince; 2, Skellefte belt; 3, Central Svecofennian subprovince; 4, Southern Svecofennian subprovince; 5, Transscandinavian igneous belt (TIB); 6, Rapakivi granites; 7, Southwest Scandinavian province; 8, Caledonides.

the Dala granitoids (and related Dala porphyries), and the Rätan granitoids. The I-type Småland-Värmland and Askersund-Finspång granitoids are mostly coarse-grained with porphyritic textures and they are dated at 1.85–1.77 Ga (U–Pb, Ahl et al. 1999; U–Pb, Persson & Wikström 1993). The Dala granitoids (including the Järna granite) range in age from 1.8 to 1.7 Ga (U–Pb) and constitute a wide variation of rocks with different textures, petrology and geochemistry (Lundqvist & Persson 1999). These rocks have undergone only low-grade metamorphism (Nyström 1983; Ahl et al. 1999). The Dala granitoids are partly covered by the Dala sandstone and the Öje basalt and have been intruded by younger dolerites (e.g. Hjelmqvist 1966; Lundqvist & Persson 1999). The early part of the Gothian orogen (1.75–1.55 Ga, Gaál & Gorbatschev 1987) coincides in time with late TIB magmatism, and these two events may thus be genetically related. Gothian rocks are mainly situated in the Southwest Scandinavian subprovince (Fig. 1), and they were reworked during the Sveconorwegian/Grenvillian orogenesis at c. 1.15–0.9 Ga (Åhäll & Gower 1997).

Geology of the Rätan granite

The c. 60×100 km^2 large, 1.70 Ga (U–Pb, Wilson et al. 1985; Patchett et al. 1987; Delin 1996) Rätan granitoid batholith is emplaced close to the present Caledonian front in western central Sweden (Fig. 2). The Rätan granite makes a significant positive anomaly in aeromagnetic data, which indicates that it carries a high content of ferromagnetic (*s.l.*) minerals. Interpretations of aeromagnetic data indicate that the pluton continues to the northwest below the Caledonides (Dyrelius 1980). This is also supported by radiometric (Rb–Sr) data from the Tännäs Augen Gneiss Nappe located northwest of the Rätan granite in the Caledonides (Claesson 1980). The Dala volcanites and associated sedimentary rocks were emplaced to the west and southwest of the Rätan batholith, and the c. 1.84–1.85 Ga Ljusdal batholith (U–Pb, Delin 1993; U–Pb, Welin et al. 1993) dominates the geology east and southeast of the pluton. Parts of the grayish and coarsely porphyritic 1.77–1.81 Ga Revsund granitoids (U–Pb, Patchett et al. 1987; Skiöld 1988; Claesson & Lundqvist 1990) are located in the northeast. In contrast with the Ljusdal batholith the Revsund and Rätan granitoids have not undergone regional metamorphism (Lundqvist et al. 1990; Delin & Aaro 1992).

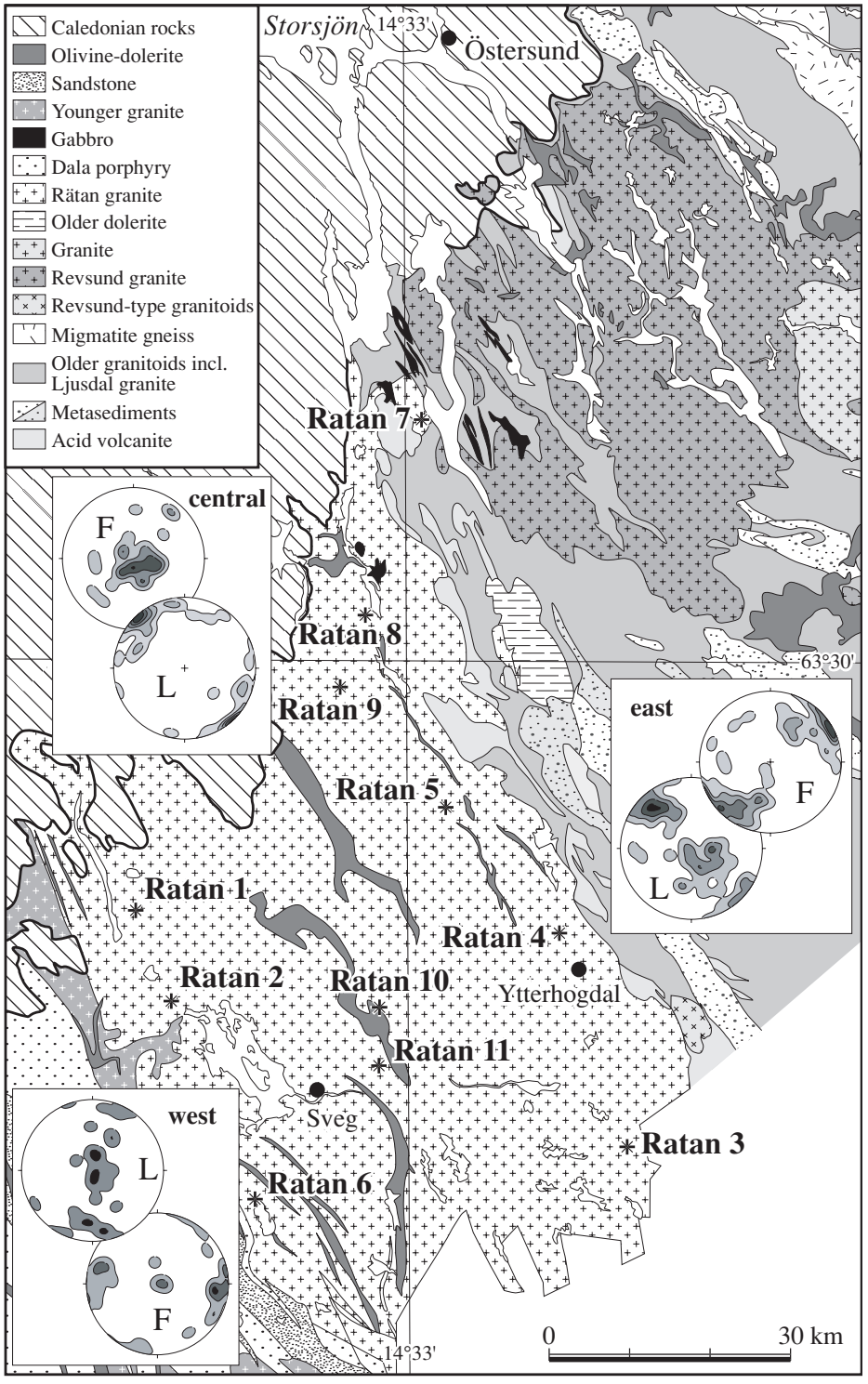


Fig. 2. Geological map of the Rätan granite and surrounding rock units (simplified from Lundegårdh et al. 1984 and Lundqvist et al. 1990). Locations of the studied sites are shown together with contour plots of directions of minimum (foliation poles - F) and maximum (lineation - L) susceptibility axes in stereographic projections for three domains; west = sites Ratan 1, 2, 6; central = sites Ratan 9–11; east = sites Ratan 3–5, 7, 8.

The Rätan granite has an almost rectangular shape and its eastern boundary coincides with a NNW trending regional deformation zone, the Storsjön-Edsbyn deformation zone (SEDZ) for approximately 100 km (Bergman & Sjöström 1994; Mattsson & Elming 2001). Unaltered olivine dolerites intrude fairly large parts of the granite. These dolerites belong to the Central Scandinavian Dolerite Group (CSDG) of post-Jotnian (c. 1.25–1.27 Ga; K–Ar, Welin & Lundqvist 1975; U–Pb, Suomi-nen 1991) dolerites that are found in the central and northern parts of Sweden and also in the southwestern parts of Finland (e.g. Gorbatschev 1979; Mattsson & Elming 2001; Elming & Mattsson 2001).

The Rätan granitoid is red in color. The texture is porphyritic and the grain size varies from fine- to coarse-grained. The rock is unfoliated and contains up to 3 cm large microcline-perthite phenocrysts, which together with quartz, biotite, hornblende and titanite dominate the mineralogy (e.g. Lundegårdh 1967; Lundqvist 1968; Ahl et al. 1997). The geochemistry of the Rätan granite is similar to that of the Småland-Värmland and Järna granitoids (Ahl et al. 1997) and it displays an alkali-rich differentiation trend, which differs significantly from the Revsund granitoids. Large numbers of mainly NE–SW and NW–SE trending brittle shear zones (some more than 20 km long), defined by low magnetic lineaments in aeromagnetic data, cross-cut the entire pluton.

Methods and sampling

Samples for paleomagnetic and AMS studies were collected from the Rätan granite at 11 sites evenly distributed across the pluton (sites Ratan 1–11, Fig. 2). A basic dyke of a width of >1 m that cuts the granite at site Ratan 6 was also sampled. A total number of 114 samples have been collected.

The samples were collected with a portable field drill, and as block samples, and orientated with a magnetic and a sun compass. Direction and intensity of the remanent magnetization were measured with a spinner magnetometer JR-5 (Geofyzika Brno) and a DC-SQUID magnetometer (2G-Enterprises). Alternating field demagnetization was performed with coils incorporated in the DC-SQUID system and thermal demagnetization was

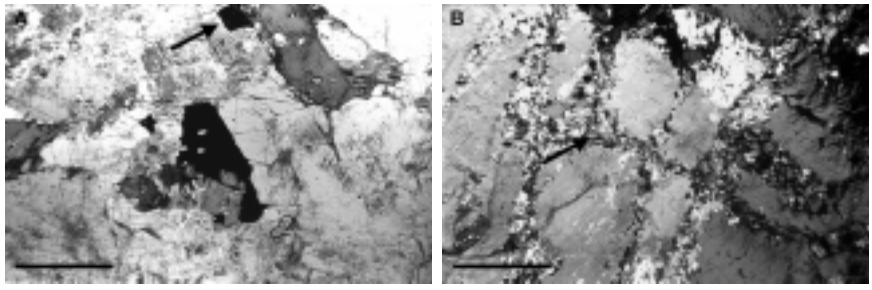


Fig. 3. Photomicrographs of thin sections of the Rätan granite, site Ratan 8 (**A**) and site Ratan 7 (**B**). The arrow in A points at a cubic magnetite grain with a biotite crystal situated close at its right. The arrow in B points at one of the fractures filled by fine-grained quartz and feldspars that cuts through the rock. Scale bars equal 2 mm.

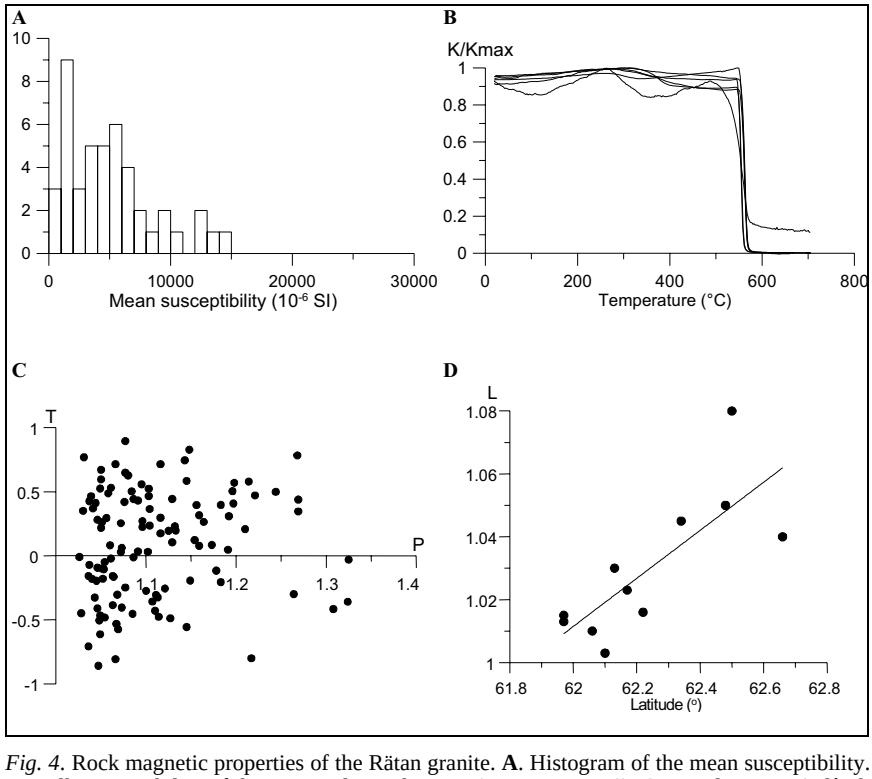


Fig. 4. Rock magnetic properties of the Rätan granite. **A**. Histogram of the mean susceptibility. **B**. Bulk susceptibility of the granite during heating (sites Ratan 1–6). **C**. P-T diagrams (Jelínek 1981). **D**. Site mean values of the lineation (sites Ratan 1–11) plotted with reference to site latitude.

performed using a Schonstedt demagnetizer TSD-1. Remanent magnetization directions were analyzed using Principle Component Analysis (Kirschvink 1980) and means were calculated using the statistics of Fisher (1953).

Anisotropy of magnetic susceptibility (AMS) was used to measure the petrofabric of rocks (e.g. Hrouda 1982). Basically, the AMS of a rock is described by a second-order tensor. This tensor can be visualized by an ellipsoid having three principal axes of maximum, intermediate and minimum susceptibility (k_{max}, k_{int} and k_{min}, respectively). Each sample from a sampling site is represented by an ellipsoid, and the ellipsoids of every sample are combined (taking into account the length and orientation of each axis respectively) to a site mean ellipsoid that represents the outcrop as a whole. For multidomain magnetite the k_{max} axis is parallel to the long axis of an ellipsoidal grain and for most rocks the k_{max} corresponds to the mineral lineation, whereas the k_{min} axis corresponds to the pole to the plane of foliation (Tarling & Hrouda 1993). $P = k_{max}/k_{min}$ characterizes the degree of anisotropy and the shape of the anisotropy ellipsoid is defined by the so-called shape parameter T (Jelínek 1981). If $-1<T<0$ the ellipsoid is prolate (cigar-shaped) and if $0<T<1$ the ellipsoid is oblate (disc-shaped); if $T=0$ the ellipsoid is neutral.

The AMS and temperature variation of magnetic susceptibility of rocks were measured with a Kappabridge KLY-3 and a CS-3 apparatus, respectively (Geofyzika Brno). The statistical analysis of the AMS data was performed using the technique of Jelínek (1978).

AMS and magmatic fabric

AMS is a well-known indicator of the direction of magma flow in basic dykes and the method has been used in several studies to define flow fabrics (e.g. Knight & Walker 1988; Bates & Mushayandebvu 1995; Raposo 1997). When magma of a basic composition crystallizes, the oxides will mimic the fabric of the pre-existing silicate phases (Tauxe et al. 1998). If this mimicking of fabrics is valid also for acid intrusions is, however, not fully understood (e.g. Borradaile & Henry 1997). Often the magnetic lineation is believed to be parallel to the magma flow direction (e.g. Bouillin et al. 1993) but lineations orientated perpendicular to the flow have also been reported (e.g. Khan 1962). Different anisotropy directions may originate from differences in the flow rate and the viscosity of a magma (Tarling & Hrouda 1993). However, according to Bouchez et al. (1990) it is accepted that the magnetic/magmatic lineation approximates the stretching axis of the finite strain ellipsoid and that a uniform orientation over large areas is a sufficient proof of shearing parallel to the lineation direction. AMS has been applied to many granite plutons for kinematic reconstruction of their emplacement (e.g. Bouillin et al. 1993 and Bouchez et al. 1990). A primary magnetic fabric can generally be related to the shape of the intrusion (Tarling & Hrouda 1993) and preferred orientations of feld-

Table 1. AMS data. n is the number of specimens, N is the number of samples, K_{mean} is the mean susceptibility (presented with its standard deviation), P is the degree of anisotropy, L is the degree of lineation and F is the degree of foliation.

Site	n	N	K_{mean}	(10^{-6} SI)	P	L	F
Ratan 1	7	7	8000	±6000	1.03	1.02	1.02
Ratan 2	9	9	9000	±9000	1.04	1.00	1.03
Ratan 3	8	8	25000	±9000	1.04	1.02	1.02
Ratan 4	8	8	33000	±7000	1.10	1.02	1.08
Ratan 5	9	9	12000	±10000	1.06	1.05	1.02
Ratan 6	11	11	5000	±3000	1.10	1.01	1.09
Ratan 7	15	9	2000	±900	1.11	1.04	1.06
Ratan 8	38	19	20000	±9000	1.17	1.08	1.09
Ratan 9	18	9	32000	±7000	1.11	1.05	1.06
Ratan 10	16	8	25000	±8000	1.04	1.03	1.01
Ratan 11	28	17	29000	±7000	1.04	1.01	1.03

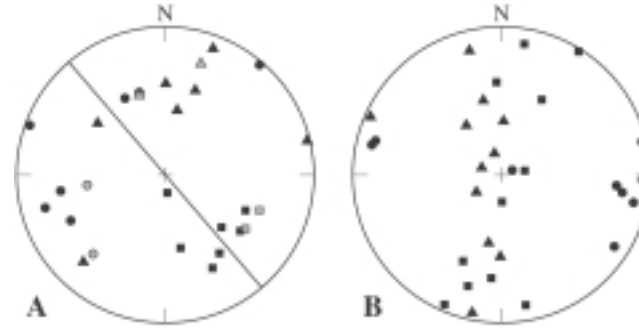


Fig. 5. Lower hemisphere equal area projection plots of AMS data for the dolerite dyke (**A**) and the Rätan granite (**B**) at the site Ratan 6. Squares/triangles/dots denote maximum/intermediate/minimum anisotropy axes, respectively. The gray symbols in A indicate the two samples collected in the 0.5 m wide altered zone along the dyke contact and the great circle in A indicates the strike and dip of the dyke plane.

spars (known as geological magma flow indicators) tend to be parallel to the magnetic foliation plane (e.g. Mattsson & Elming 2001). If the magnetic foliation of a granite is parallel to a regional foliation, the pattern may be due to large scale deformation.

Results

Petrography

At none of the sites in the Rätan granite there is evidence of plastic deformation. At the site Ratan 8 (Fig. 2) the rock has a heterogeneous grain-size distribution with fine-grained parts distinctly separated from coarse-grained. In some parts the rock has a basic composition. Microcline-perthitic phenocrysts, biotite, quartz and plagioclase dominate the mineralogy, but zircons and epidote are also seen. Magnetite grains are cubic in shape, which indicates a primary origin (Fig. 3A). Minor cubic grains of hematite are found and their shapes indicate that they may have been formed as a result of oxidation of magnetite. Hematite is also often associated with magnetite.

At site Ratan 10, in the central part of the granite (Fig. 2), the rock has suffered from minor chlorite alteration. At Ratan 7, which is situated at the northeastern boundary of the pluton and close to the deformation zone (SEDZ), the rock has fractures filled by fine-grained quartz and feldspars (Fig. 3B). This clearly demonstrates crushing of the rock and indicates that the granite has suffered from low-grade alteration and brittle deformation. At this site the rock also carries mica-crystals with a "bottle-green" color that contain lamellae of ilmenite, which may indicate a metamorphic grade that corresponds to prehnite–pumpellyite-facies.

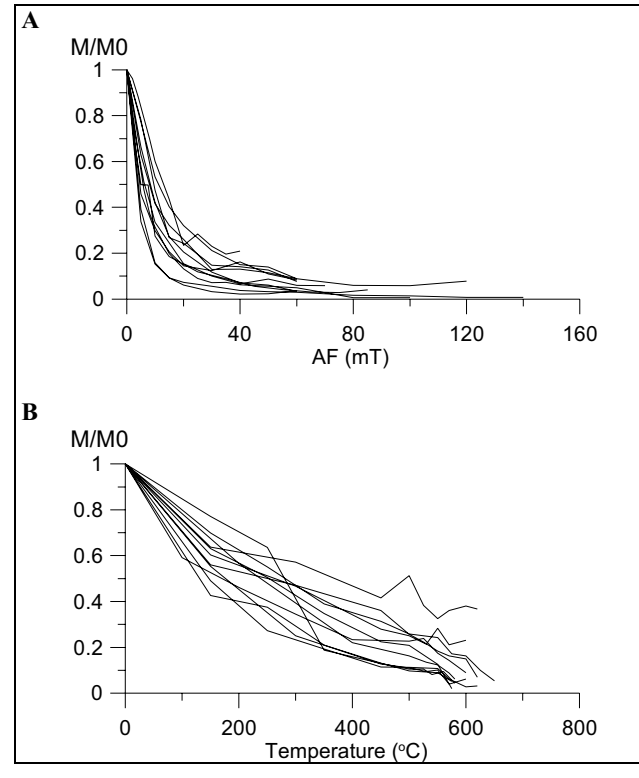


Fig. 6. Examples of alternating field demagnetization curves (**A**) and thermal demagnetization curves (**B**) for one sample from each of the sites Ratan 1–11 of the Rätan granite.

AMS

The magnetic fabric of the Rätan granite varies systematically across the pluton. The five sites Ratan 3–5, 7, 8 have consistent directions of the anisotropy axes with subvertical foliation planes trending NW–SE, parallel to the eastern boundary of the granite (Fig. 2). The lineations define a girdle pattern indicating a rotation of grains. Along the western boundary (sites Ratan 1, 2, 6) we find a coherent magnetic fabric with mainly subvertical magnetic foliation planes trending N–S and a girdle distribution of the lineations (Fig. 2). The strike of the foliation planes of the sites Ratan 2 and 6 makes an angle of c. 20–30° with the trend of the western boundary of the granite, whereas the strike of the foliation planes of Ratan 1 is NW–SE, which is parallel to the contact. Three sites in the center of the pluton (Ratan 9–11) carry AMS directions that are significantly different from the other sites (also in Mattsson & Elming 2001). The foliation planes are subhorizontal and the lineations cluster in a NW–SE trending di-

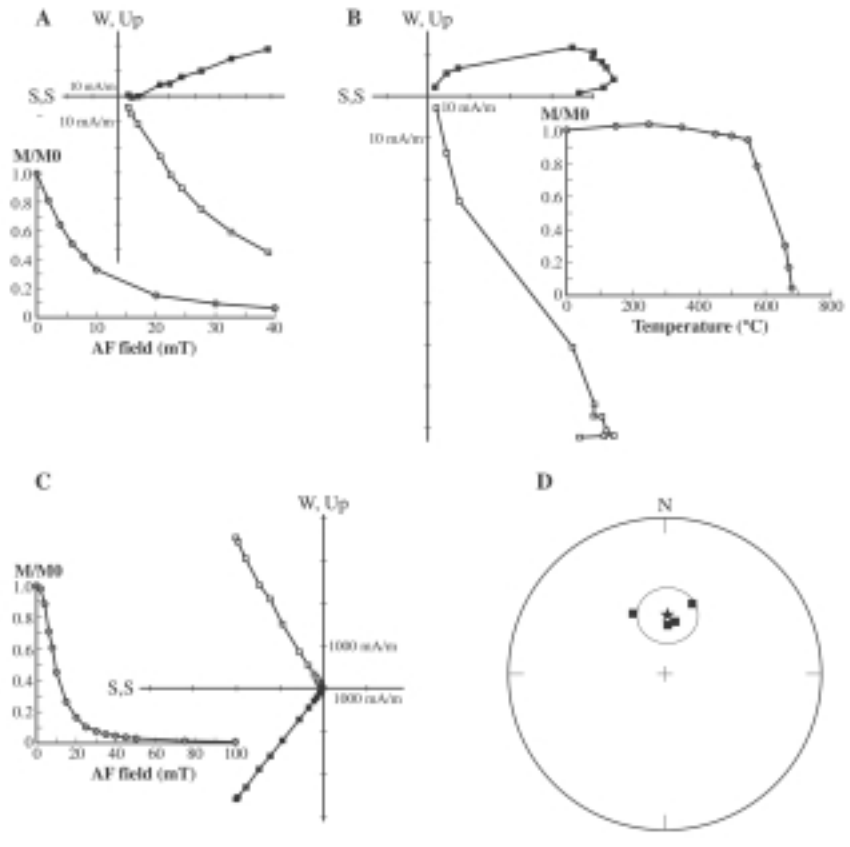


Fig. 7. Examples of remanence behavior during alternating field (**A**) and thermal (**B**) demagnetizations. An example of a reversed direction of magnetization at the site Ratan 8 (**C**). Open (closed) symbols in the vector plots show vertical (horizontal) projection. In the lower hemisphere equal area projection plot the formation mean remanence direction (star) and its 95% cone of confidence of the Rätan granite is shown (**D**). This mean is based on the four site mean directions (squares) of the sites Ratan 5, 6, 8 and 10.

rection (Fig. 2). This indicate that the central part of the granite body has a subhorizontal maximum finite strain striking NW–SE.

The Rätan granite is highly magnetic (Table 1) and the average volume susceptibility of the entire pluton is 0.019±0.013 SI (Fig. 4A), which suggests that the rock is an I-type or A-type granite (Tarling & Hrouda 1993). Along the eastern and western boundaries the mean susceptibility is 0.018±0.012 SI (sites Ratan 3–5, 7, 8) and 0.007±0.008 SI (sites Ratan 1, 2, 6), respectively, which is lower than in the central part where the mean is 0.030±0.007 SI (sites Ratan 9–11). Rock magnetic data show that magnetite completely governs the bulk susceptibility of the granite (Fig. 4B).

The average degree of anisotropy of the granite is P = 1.11±0.07 and there is an even distribution of prolate and oblate shapes of the ellipsoids (Fig. 4C). The degrees of foliation (F) and lineation (L) are below 10% for site mean ellipsoids (Table 1) and there is a trend of successively increasing degrees of lineation moving from south to north in the pluton (Fig. 4D).

The dolerite dyke truncating the granite at site Ratan 6 carries a mean susceptibility of 0.0010±0.0002 SI. Rock magnetic data indicate that magnetite governs the susceptibility of the dyke and in some of the drill cores the rock has a greenish color indicating that it may have suffered from chlorite alteration. The granite, however, appears to be unaltered. Minimum and intermediate anisotropy axes create a girdle distribution, whereas the maximum axes are clustered in the dyke plane, which strikes NW–SE and dips subvertically (Fig. 5A). This AMS pattern results in a strongly prolate site mean anisotropy ellipsoid and it indicates that the grains underwent rotation during the magma flow (Knight & Walker 1988). The anisotropy parameters are poorly developed and hardly reach above 1%. This may indicate that the dyke carries a primary flow fabric, but most likely also reflects the low content of ferromagnetic (*s.l.*) minerals as indicated by the low mean susceptibility. Two samples were collected in a c. 0.5 m wide zone of altered rocks (with a mainly basic composition) along the western margin of the dyke. The orientation of the anisotropy axes and the mean susceptibility of these samples correlate well with the dyke and deviate significantly from those of the granite (Fig. 5B).

Paleomagnetism

The Rätan granite. – Alternating field (AF) demagnetization was performed in fields up to 60–160 mT. The intensity of magnetization generally decreases exponentially (Fig. 6A) and the coercivity of the natural remanent magnetization (NRM) is most often low, with an average median destructive field (MDF; Dunlop & Özdemir 1997) of 8 mT. Sometimes it is not possible to completely demagnetize the samples and there are indications of overlapping coercivity spectra. This suggests that the carrier of remanence is predominantly multidomian magnetite and that there is a small amount of grains with much higher coercivity, e.g., single-domian magnetite, hematite or pyrrhotite, present in the samples. The low coercivity directions show steep inclinations and scattered declinations, though often close to the present earth magnetic field (PEF), which indicates a viscous origin of the magnetization (VRM). Vector plots of the AF directions usually reveal one or two components (Fig. 7A) and the high coercivity components are sometimes unstable. Linear vector fits straight to origo can be defined for several samples at the sites Ratan 2, 6, 8 and 11, but are otherwise rare.

During thermal demagnetization most samples lose more than 50% of their initial magnetization intensity at 300°C, and often there is a linear decrease of the intensity from 20°C to 600–650°C, which indicates a wide range of unblocking temperatures (Fig. 6B). The thermal demagnetization curves indicate the presence of both magnetite (or titanomagnetite) and

hematite in the granite. Low unblocking temperature components (T<400°C) of magnetization can be isolated at all sites except at Ratan 3, and the defined directions are generally close to the PEF. High temperature components (T>450°C) can be defined only in rare cases, except at the site Ratan 8 where nine samples carry a stable NRM up to 600–680°C (Fig. 7B) and linear vector fits can be performed straight to origo. Vector directions defined above temperatures of 575°C do not differ from those defined in the interval 500–575°C. At site Ratan 8 one sample carries a stable NRM in high AF fields (20–100 mT) as well as at high temperatures (500–575°C) with the direction of decl. = 129° and incl. = –46° (Fig. 7C). The direction is of opposite polarity to the majority of directions and it most likely indicates that the magma cooled during a period of a magnetic polar reversal (McElhinny 1973). An origin of magnetic self-reversal can however not be excluded (Dunlop & Özdemir 1997).

The site mean directions are poorly defined with rather high α95 values and precision parameters (Fisher 1958) around 10–40 (Table 2). A formation mean direction is defined from the site means Ratan 5, 6, 8 and 10 for which the α95<20° (Table 2). The calculated mean has decl. = 2°, incl. = 59°, α95 = 15° and k = 37 (Fig. 7D), which yields a paleopole position of Plat = 67° and Plon = 190° (dp = 17°, dm = 23°) (Fig. 10).

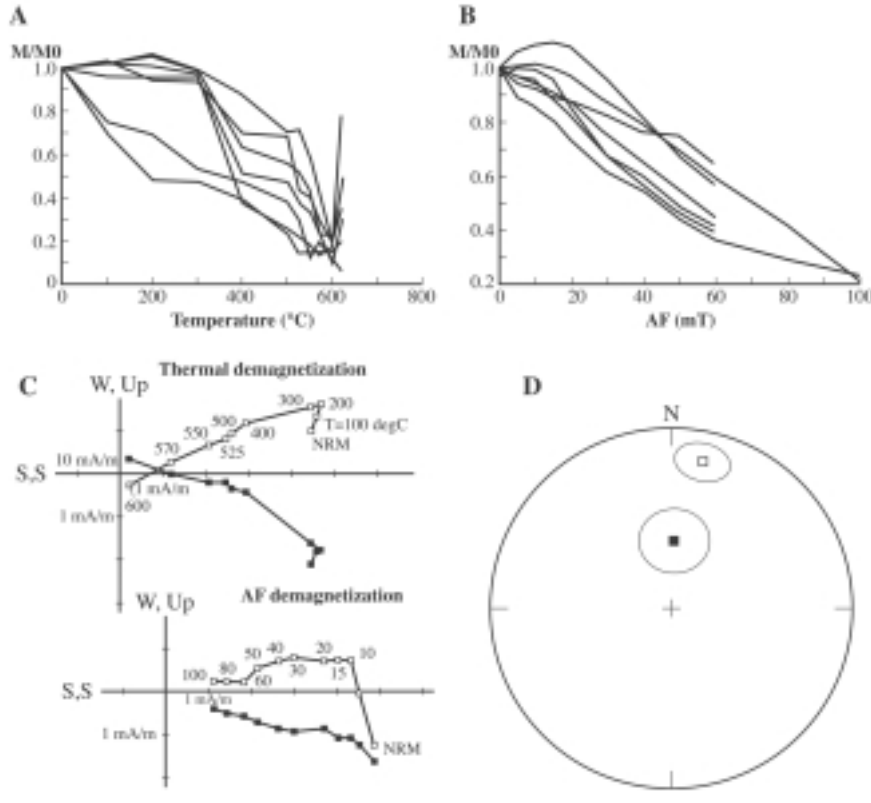


Fig. 8. Paleomagnetic results from the site Ratan 6. Thermal (**A**) and alternating field (**B**) demagnetization curves for seven samples from the dolerite dyke. Examples of vector plots from thermal and AF demagnetizations of the dyke (**C**). Open (closed) symbols in the vector plots show vertical (horizontal) projection. Lower hemisphere equal area projection plot of the site mean remanence directions of the dolerite dyke (open square) and the Rätan granite (closed square) (**D**). The open square in D indicates a negative inclination and the ellipses show the 95% cone of confidence of the mean directions.

Table 2. Paleomagnetic data. Glat/Glon is the geographic latitude and longitude, N is the number of samples, Decl. (Incl.) is the declination (inclination) of the remanent magnetization, k is the best estimate of the Fisher precision parameter, α95 is the radius of the 95% cone of confidence, Plat (Plon) is the latitude (longitude) of the virtual geomagnetic pole position and dp/dm are the semi-axes of the 95% confidence ellipsoid of the geomagnetic pole position.

Site	Glat/Glon	N	Decl.	Incl.	k	α95	Plat.	Plon.	dp/dm
Ratan 1	62.22/13.92	4	23	72	18	22	77	120	34/39
Ratan 2	62.10/14.00	5	—	—	—	—	—	—	—
Ratan 3	61.97/14.92	2	—	—	—	—	—	—	—
Ratan 4	62.17/14.87	3	—	—	—	—	—	—	—
Ratan 5	62.34/14.65	4	21	50	35	16	56	161	14/21
Ratan 6 (granite)	61.97/14.17	8	11	62	15	15	70	170	17/23
Ratan 6 (dyke)	61.97/14.17	7	12	–18	37	10	18	182	5/11
Ratan 7	62.66/14.63	—	—	—	—	—	—	—	—
Ratan 8	62.50/14.50	11	333	54	20	11	57	239	10/15
Ratan 9	62.48/14.39	6	335	68	9	24	72	255	34/41
Ratan 10	62.13/14.50	7	3	64	16	16	73	186	20/25
Ratan 11	62.06/14.50	17	—	—	—	—	—	—	—
Formation mean (sites 5, 6, 8, 10)	62.24/14.46	30	2	59	37	15	67	190	17/23

Baked contact tests. – The intrusion of a dyke into an older rock causes a reheating of the host rock. Depending on the degree of heating and the magnetic properties of the host rock, and under the assumption that no chemical alteration occurs, the host rock will be thermally re-magnetized. During cooling of the dyke the re-heated host rock may acquire a partial thermoremanent magnetization (pTRM) with a direction parallel to the earth magnetic field at the time of intrusion of the dyke. The host rock at 20–30 m distance from the intruding dyke retains its original magnetization. If we obtain significantly different characteristic magnetizations of the re-heated host rock and the host rock at far distance, this baked contact test is positive. A positive baked contact test provides indirect evidence that the magnetization of the dyke and the host rock has been stable since the time of the intrusion (McElhinny 1973).

Thermal demagnetization of samples from the dolerite dyke that cuts the Rätan granite at site Ratan 6 indicates that magnetite (or titanomagnetite), hematite and pyrrhotite carry the NRM of the rock (Fig. 8A). The large increases in magnetization at temperatures above 600°C probably indicate changes in magnetic mineralogy caused by the heating and cooling of the specimens in air (Fig. 8A). Low unblock-

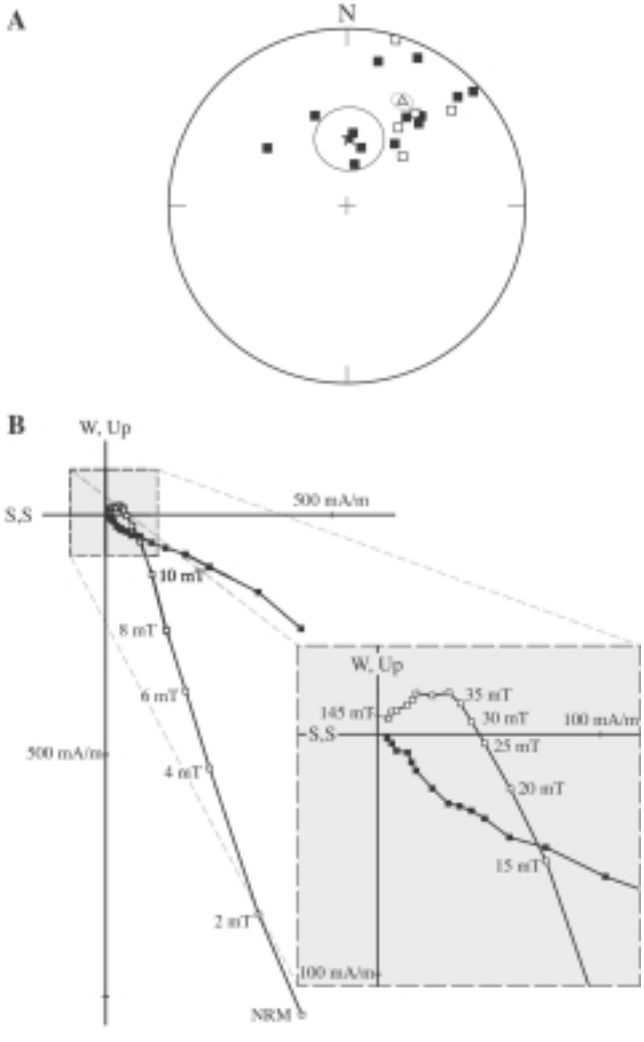


Fig. 9. Paleomagnetic results from the site Ratan 11. **A**. Plot of the high coercivity remanence directions of the granite together with the formation mean of the Rätan granite (star) and the site mean direction of the dolerite sill (triangle, Mattsson & Elming, 2001). The open symbols indicate negative inclination and the ellipses show the 95% cone of confidence of the mean directions. **B**. Example of a vector plot from an alternating field demagnetization of a granite sample situated close to the contact of the sill. Open (closed) symbols in the vector plot show vertical (horizontal) projection.

ing temperature components (T<200°C) yield magnetization directions with steep inclination and scattered declinations, probably indicating a VRM influenced by the PEF. A large drop in magnetization intensity at T = 300–400°C (Fig. 8A) is observed for four samples and may reflect the Curie temperature for pyrrhotite. For three of these samples a mean direction of decl. = 37°, incl. = -10°, $\alpha 95$ = 4° and k = 824 is defined (T = 200–400°C, Mean angular deviation MAD <5°). At temperatures of T = 500–570°C vector directions with NNE declinations and a sub-horizontal inclination are well defined (Fig. 8C), and these directions are also defined above the Curie temperature for magnetite. AF demagnetization of dyke samples (Fig. 8B) shows very high coercivities (MDF = 45mT; Dunlop & Özdemir 1997) and indications of overlapping coer-civity spectra. Low coercivity components with directions close to the PEF indicate a VRM origin. In high AF-fields (20–100 mT) the vectors of magnetization decline towards the origo. These vector directions correlate well with the high unblocking temperature directions (Fig. 8C). The site mean remanence direction of the dyke (decl. = 12°, incl. = -18°, k = 37 and $\alpha 95$ = 10°; Table 2) is significantly different from the site mean of the Rätan granite (decl. = 11°, incl. = 58°, $\alpha 95$ = 18°; Table 2) (Fig. 8D). A granite sample, which was collected at 1.4 m distance from the contact, carries a stable remanence direction very close to that of the dyke. The direction is defined in a high coercivity interval (AF field = 60–80mT) and the vector declines towards the origo during demagnetization, which indicates that the granite is completely remagnetized at this distance from the contact. These results suggest that the baked contact test for the dyke is positive.

The site Ratan 11 is situated close to an outcropping dolerite sill with a characteristic TRM of decl. = 28°, incl. = –34°, $\alpha 95$ = 4° and k = 77 (Mattsson & Elming 2001). Samples from the Rätan granite were here collected in two outcrops, c. 30 m northwest of the sill and c. 200 m north of the sill. The thermal data are very poor, and no high unblocking temperature vectors can be isolated. The directions of the high coercivity components are distributed along a great circle, with some directions close to the formation mean of the Rätan granite while others are close to the site mean direction of the dolerite (Fig. 9A). The directions between these two mean values indicate mixed remanence components. The sill directions in the granite are isolated in various coercivity intervals in the range of 20mT to 145mT, and the vector directions decline to origo (Fig. 9B). These directions are found in both outcrops, close to samples that carry the characteristic granite direction, and the coercivity spectra of both granite and sill directions overlap. The overlapping coercivity intervals in which sill and granite directions are isolated probably indicate that a chemical alteration rather than a partial re-heating caused the re-magnetization of the granite. However, it is also possible that both thermal and chemical re-magnetization followed due to the intrusion of the dolerite sill.

Discussion

The petrographic study indicates that the magnetite grains in the Rätan granite are primary, and the thin sections reveal only weak brittle deformation and low-grade alteration of the rock. The average degree of magnetic anisotropy of the granite is P = 1.11. Plutonic rocks with a primary magnetic fabric have P-values of c. 1.10 or less (Hrouda 1982) and plastically deformed rocks with such high mean susceptibility as the Rätan granite are expected to have significantly higher degree of anisotropy, sometimes up to P = 2.0–3.0 (Mattsson & Elming 2001). These results, and the fact that the anisotropy pattern broadly follows the shape of the pluton, indicate that the granite carries a primary magnetic/magmatic fabric. The measurements of the magnetic properties of the granite indicate that magnetite or titanomagnetite governs the AMS. We suggest that the magnetic fabric reflects the different stress conditions that prevailed during a late stage of the emplacement of the magma and that the magnetic anisotropy orientation is related to the flow direction.

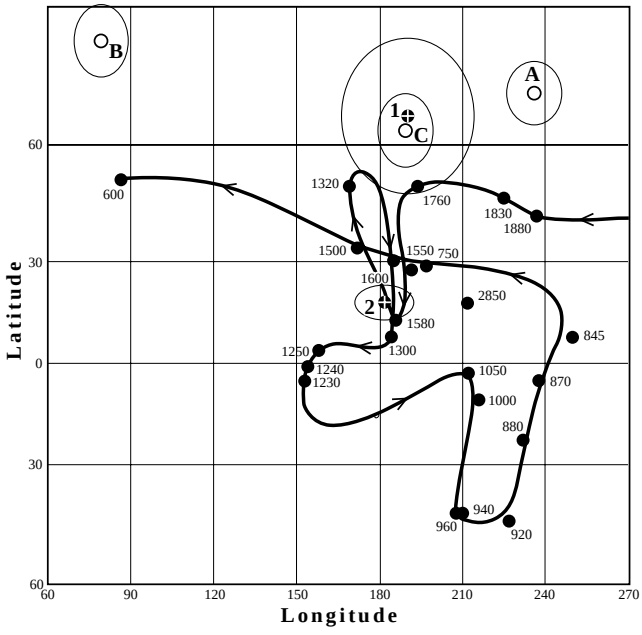


Fig. 10. The Apparent Polar Wander Path (APWP) for the Fennoscandian Shield (after Elming et al. 1993). Closed symbols along the APWP denote successive Grand Mean Poles. Symbols with a white cross denote paleopoles of the Rätan granite (1) and the dyke at site Ratan 6 (2) from this study. Open symbols denote paleopoles from Bylund & Elming (1992), with notation according to Table 3.

Table 3. Paleomagnetic data from Bylund & Elming (1992). See Table 2 for explanation.

Site	Glat/Glon	N	Decl.	Incl.	k	α95	Plat.	Plon.	dp/dm
Blyberget (A)	61.16/14.16	5	343	65	106	6	73	236	8/10
Hållan (B)	61.13/14.11	6	15	76	42	4	83	79	7/8
Dala Floda (C)	60.53/14.75	9	3	54	42	7	64	189	7/10

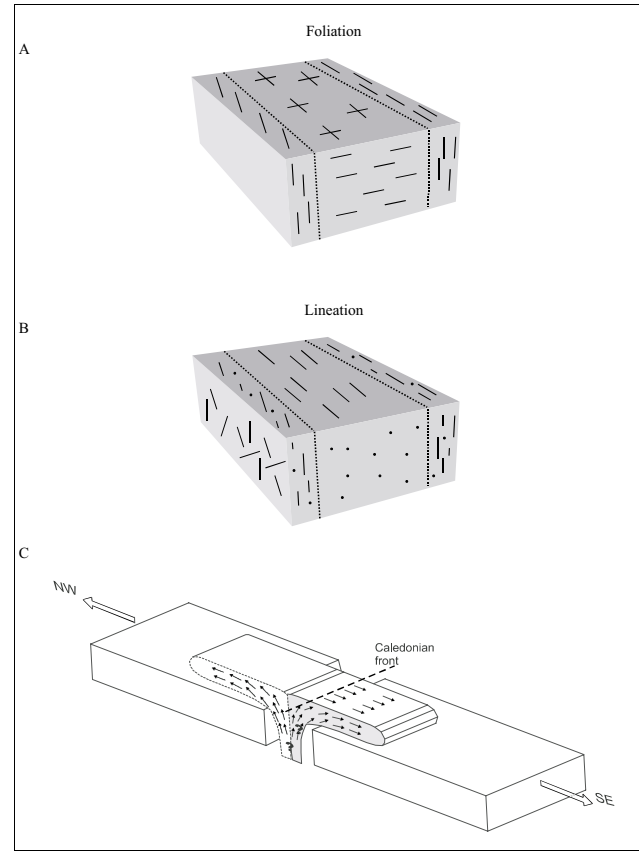


Fig. 11. Block diagrams (not to scale) illustrating the foliation (**A**) and lineation (**B**) pattern of the three contrasting magnetic fabric domains in the Rätan granite. Schematic diagram (not to scale) illustrating the proposed scenario of the emplacement of the Rätan granite (**C**). A possible northwestward continuation of the Rätan granite, as drawn in the figure, is indicated in aeromagnetic data (Dyrelius 1980).

The present AMS study of the Rätan granite identified three contrasting magnetic fabric domains: the eastern (sites Ratan 3–5, 7, 8), the central (sites Ratan 9–11) and the western domain (sites Ratan 1, 2, 6; Fig. 2). The width of the eastern and western domains is inferred at 10–12 km (measured perpendicularly from the eastern/western contact, respectively), which together represents c. 40% of the total width of the pluton. The western domain may possibly be wider, while the site Ratan 9 situated c. 12 km west of the eastern contact restricts the width of the eastern domain to c. 12 km. The width of the central domain is at least c. 20 km, which is the distance between the sites Ratan 9 and 11 measured along a line perpendicular to the eastern and western contacts, but could be as much as 30–40 km.

The eastern and western domains are dominated by subvertical magnetic foliation planes and a girdle distribution of the maximum (lineation) and intermediate anisotropy axes (Fig. 11A–B). This pattern indicates a magma flow parallel to the contacts and dominant subhorizontal NE–SW (in the E) and E–W (in the W) directed compressive deformation together with a simple shear deformation causing a rotation of the grains. The c. 20–30° angle between the trend of the western contact and the strike of the magnetic foliation planes could possibly be a result of sinistral oblique shearing, i.e. transpression, along the western contact zone. The mean susceptibility is significantly lower in the western domain compared to the central domain, which indicates a lower content of iron in this part of the granite.

In the central domain subhorizontal foliation planes and the cluster of the lineation in a NW–SE direction may indicate a subhorizontal flow of magma in that same direction (Fig. 11A–B). This suggests that the pluton was fed obliquely with a magma source to the present southeast or northwest. Under the assumption that the degree of strain is highest close to the magma source (where there is likely to be a concentration of stress), the trend of decreasing degree of lineation with lower latitudes may indicate a southward direction of the flow. Since c. 60% of the width of the pluton (in the east–west direction) appears to be dominated by a subhorizontal magnetic fabric, we suggest that the Rätan granite was emplaced in a tectonic regime governed by NW–SE directed extension and that the root is located to the present northwest (Fig. 11C). This supports the model proposed by Ahl

et al. (1999), which suggests that the 1.68–1.70 Ga Dala granitoids were emplaced in an extensive tectonic environment. The AMS fabric in the eastern and western domains is believed to reflect the stress field along the contacts, which should give rise to a pattern of foliations parallel to the boundary of the pluton (Tarling & Hrouda 1993). The absence of foliation planes with a moderate dip, which one would expect in a transition zone between the boundaries and the central domain, is probably explained by the relatively large spacing (5–20 km) between the sampling sites.

If the formation mean direction of the Rätan granite (decl. = 2°, incl. = 58°, $\alpha 95$ = 15° and k = 38) is carried by a primary TRM the result provides important new information to the present Fennoscandian apparent polar wander path (APWP; Elming et al. 1993). However, the poor results from thermal demagnetization and the fact that the established formation mean direction is not significantly different from the direction of the PEF argue against a primary origin of the magnetization. Due to the fact that multi-domain magnetite grains tend to govern the remanent magnetization in highly magnetic granitoids (Tarling & Hrouda 1993), the similarity to the PEF may indicate a viscous origin. This is, however, not likely to be the case for the Rätan granite. Though overlapping coercivity spectra do occur, we have been able to define stable directions in AF fields well above 100mT and at temperatures above 550°C, which ought to be enough to erase any viscous component. A reversed magnetization identified at the site Ratan 8 indicates that the magma may have cooled during a period of a magnetic polar reversal (McElhinny 1973), which may support a primary origin of the magnetization. A positive baked contact test has been demonstrated for a dolerite dyke at the site Ratan 6. The paleopole (Plat = 18°, Plon = 182°, dp = 5° and dm = 11°) calculated from the mean direction of the dyke corresponds to two possible paleomagnetic age intervals of c. 1.30–1.32 and c. 1.5–1.6 Ga (Fig. 10). A pole position of similar direction (Plat = 25°, Plon = 190°) was found in a NE–SW trending basic dyke cutting the Storsjön-Edsbyn deformation zone (SEDZ, Fig. 1) only c. 10 km east of the eastern boundary of the Rätan granite (Mattsson & Elming 2001). Similar pole positions were also reported by Moakhar & Elming (1998) from basic dykes in the Åsele region (c. 100 km north of the Rätan granite), from the so-called E–W dykes in central Sweden (Bylund 1985) and from the Föglö – Sottunga dykes in the Åland archipelago (Pesonen & Neuvonen 1981). The Föglö – Sottunga dykes are dated at 1540±12 Ma (U–Pb, Pesonen & Neuvonen 1981). It is therefore possible that the dyke at site Ratan 6 should be placed in the same time interval. If so, the result indicates that the magnetization of the Rätan granite at the site Ratan 6 has been stable at least since c. 1.5–1.6 Ga. A chemical and/or thermal re-magnetization of the Rätan granite has also been demonstrated at the site Ratan 11, where a dolerite sill (site ST15 in Mattsson & Elming 2001) intrudes the granite. The characteristic remanence direction of the sill (decl. = 28°, incl. = –34°) is similar to directions that have been isolated in more than 50 sites of post-Jotnian dolerite sills that are widely distributed in the Fennoscandian Shield, including in the Rätan granite (Elming & Mattsson 2001). It is therefore likely that the sill at site Ratan 11 belongs to this formation of post-Jotnian sills and that the intrusion of the sill gave rise to the chemical re-magnetization of the granite as indicated in this study. This contact test is not strictly positive (McElhinny 1973), but it may indicate that the magnetization of the sill and the Rätan granite has been stable since the intrusion of the sill at c. 1.25 Ga (Elming & Mattsson 2001).

The paleomagnetic pole position (Plat = 66°, Plon = 190°, dp = 16° and dm = 22°) calculated for the Rätan granite indicates that the Fennoscandian Shield was situated at a slightly higher latitudinal position at c. 1.70 Ga than at present. The longitude of the pole position broadly correlates with the 1.60–1.76 Ga reference poles of the APWP for the Fennoscandian Shield (Elming et al. 1993) (Fig. 10), but it is clearly in a higher latitude. From fairly poor paleomagnetic data from the Dala porphyries and granitoids (Bylund & Elming 1992) poles have been calculated that some are close to the pole of the Rätan granite (Table 3 and Fig. 10). Lundqvist & Persson (1999) present new radiometric data (U–Pb) of these rocks that indicate an age of c. 1.70 Ga, which is close in time to the age of the Rätan granite (1.70 Ga, U–Pb, Wilson et al. 1985; Patchett et al. 1987; Delin 1996). All these data argue for a primary origin of the remanent magnetization of the Rätan granite.

Conclusions

Magnetite or titano-magnetite primarily governs the magnetic susceptibility of the Rätan granite, whereas multi-domain magnetite, but also single-domain magnetite and hematite mainly govern the remanent magnetization. The magnetic fabric of the granite is of primary origin and reflects the different stress conditions that prevailed during a late stage of the emplacement of the magma, which suggests that the magnetic anisotropy orientation is related to the flow direction. Three contrasting magnetic fabric domains are identified. The eastern and western domains are c. 12 km wide and the magnetic fabric is characterized by NW–SE to N–S trending subvertical foliation planes and a girdle distribution of the maximum and intermediate anisotropy axes. The central domain is c. 20–40 km wide and it is characterized by subhorizontal magnetic foliation planes and a clustering of the lineation in a NW–SE trending direction. We conclude that the Rätan granite was emplaced in a tectonic regime governed by NW–SE directed extension and that the pluton was fed obliquely with a magma source situated to the present northwest.

A positive baked contact test for a basic dyke and a magnetization direction of reversed polarity of the Rätan granite support a primary origin of the remanent magnetization of the granite. From the paleomagnetic formation mean direction of decl. = 2°, incl. = 59° ($\alpha 95$ = 15° and k = 37) a paleopole position (Plat = 67°, Plon = 190°, dp = 17° and dm = 23°) is calculated that corresponds to an age of c. 1.7 Ga.

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References

Åhäll, K.-I. & Gower, C.F., 1997: The Gothian and Labradorian orogens: Variations in accretionary tectonism along a late Palaeoproterozoic Laurentia–Baltica margin. *GFF 119*, 181–191

Åhäll, K.-I. & Larson, S.Å., 2000: Growth related 1.85–1.55 Ga magmatism in the Baltic Shield; a review addressing the tectonic characteristics of Svecofennian, TIB 1-related, and Gothian events. *GFF 122*, 193–206.

Ahl, M., Andersson, U.B., Lundqvist, T. & Sundblad, K., 1997: Rapakivi granites and related rocks in central Sweden. *Sveriges Geologiska Undersökning Ca 87*, 1–99.

Ahl, M., Sundblad, K. & Schöberg, H., 1999: Geology, geochemistry, age and geotectonic evolution of the Dala granitoids, central Sweden. *Precambrian Research 95*, 147–166.

Andersson, U.B., 1991: Granitoid episodes and mafic-felsic magma interaction in the Svecofennian of the Fennoscandian shield, with main emphasis on the ≈ 1.8 Ga plutonics. *Precambrian Research 51*, 127–149.

Bates, M.P. & Mushayandebvu, M.F., 1995: Magnetic fabric in the Umvimeela Dyke, satellite of the Great Dyke, Zimbabwe. *Tectonophysics 242*, 241–254.
[Bergman, S. & Sjöström, H., 1994: *The Storsjön-Edsbyn Deformation Zone, central Sweden*. Research report, Geological Survey of Sweden. 46 pp.]
Borradaile, G.J. & Henry, B., 1997: Tectonic applications of magnetic susceptibility and its anisotropy. *Earth Science Reviews 42*, 49–93.
Bouchez, J.L., Gleizes, G., Djouadi, T. & Rochette, P., 1990: Microstructure and magnetic susceptibility applied to emplacement kinematics of granites: the example of the Foix pluton (French Pyrenees). *Tectonophysics 184*, 157–171.
Bouillin, J.-P., Bouchez, J.-L., Lespinasse, P. & Pêcher, A., 1993: Granite emplacement in an extensional setting: an AMS study of the magmatic structures of Monte Capanne (Elba, Italy). *Earth and Planetary Science Letters 118*, 263–279.
Bylund, G., 1985: Paleomagnetism of middle Proterozoic basic intrusives in central Sweden and the Fennoscandian apparent polar wander path. *Precambrian Research 28*, 283–310.
Bylund, G. & Elming, S.-Å., 1992: The Dala dolerites, central Sweden, and their paleomagnetic signature. *Geologiska Föreningens i Stockholm Förhandlingar 114*, 143–153.
Claesson, S., 1980: A Rb–Sr isotope study of granitoids and related mylonites in the Tännäs Augen Gneiss Nappe, southern Swedish Caledonides. *Geologiska Föreningens i Stockholm Förhandlingar 102*, 403–420.
Claesson, S. & Lundqvist, T., 1990: Svecofennian granites in the Bothnian Basin, central Sweden. (Abstract, 19:e Nordiske Geologiske Vintermøte, Stavanger, Norway.) *Geo-nytt 1/90*, 36.
Delin, H., 1993: The radiometric age of the Ljusdal granodiorite of central Sweden. *Sveriges Geologiska Undersökning C 823*, 13-16.
Delin, H., 1996: U-Pb ages of granitoids in the Kårböle region, central Sweden. *Sveriges Geologiska Undersökning C 828*, 6–14.
Delin, H. & Aaro, S., 1992: Berggrundskartorna 16F Kårböle, 1:50,000. *Sveriges Geologiska Undersökning Ai*, 56-59.
Dunlop, D.J. & Özdemir, Ö., 1997: *Rock magnetism. Fundamentals and frontiers*. Cambridge University Press. 573 pp.
Dyrelius, D., 1980: Aeromagnetic interpretation in a geotraverse area across the central Scandinavian Caledonides. *Geologiska Föreningens i Stockholm Förhandlingar 102*, 421–438.
Elming, S.-Å. & Mattsson, H., 2001: Post Jotnian basic intrusions in the Fennoscandian Shield, and the break up of Baltica from Laurentia: a palaeomagnetic and AMS study. *Precambrian Research 108*, 215–236.
Elming, S.-Å., Pesonen, L.J., Leino, M.A.H., Khramov, A.N., Mikhailova, N.P., Krasnova, A.F., Mertanen, S., Bylund, G. & Terho, M., 1993: The drift of the Fennoscandian and Ukranian shields during the Precambrian: a paleomagnetic analysis. *Tectonophysics 223*, 177–198.
Fisher, R.A., 1953: Dispersion on a sphere. *Proceedings of the Royal Society London A217*, 295–305.
Fredén, C. (ed.), 1994: *Geology*. 208 pp. National Atlas of Sweden. SNA Publishing.
Gaál, G. & Gorbatschev, R., 1987: An outline of the Precambrian evolution of the Baltic Shield. *Precambrain Research 35*, 15–52.
Gorbatschev, R. & Bogdanova, S., 1993: Frontiers in the Baltic Shield. *Precambrain Research 64*, 3–21.
Gorbatschev, R., Solyom, Z. & Johansson, I., 1979: The central Scandinavian Dolerite Group in Jämtland, central Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 101*, 177–190.
Hjelmqvist, S., 1966 Beskrivning till berggrundskarta över Kopparbergs län. *Sveriges Geologiska Undersökning Ca 40*, 1–217.
Högdahl, K. & Sjöström, H., 1999. Timing and kinetmatics of Paleoproterozoic shear zones in central Sweden. *European Union of Geologists Meeting 10, Strasbourg, France*, 828.
Hrouda, F., 1982: Magnetic anisotropy of rocks and its application in geology and geophysics. *Geophysical Surveys 5*, 37–82.
Jelínek, V., 1978: Statistical processing of anisotropy of magnetic susceptibility measured on groups of specimens. *Studia Geophysica et Geodaetica 22*, 50–62.
Jelínek, V., 1981: Characterisation of the magnetic fabric of rocks. *Tectonophysics 79*, 63–67.
Khan, M.A., 1962: The anisotropy of magnetic susceptibility of some igneous and metamorphic rocks. *Journal of Geophysical Research 67*, 2873–2885.
Kirschvink, J.L., 1980: The least-squares line and plane and the analysis of paleomagnetic data. *Geophysical Journal of the Royal Astronomical Society 62*, 699–718.
Knight, M.D. & Walker, G.P.L., 1988: Magma flow directions in dikes of the Koolau Complex, Oahu, determined from magnetic fabric studies. *Journal of Geophysical Research 93*, 4301–4319.
Larson, S.-Å. & Berglund, J., 1992: A chronological division of the Transscandinavian Igneous Belt – three magmatic episodes? *Geologiska Föreningens i Stockholm Förhandlingar 114*, 459–460.
Lundegårdh, P.H., 1967: Berggrunden i Gävleborgs län. *Sveriges Geologiska Undersökning Ba 22*, 1–303.
Lundegårdh, P.H., Strömberg, A., Karis, L., Zachrisson, E., Sjöstrand, T., Skoglund, R., Gorbatschev, R. & Kornfält, K.-A., 1984: Karta över berggrunden i Jämtlands län. *Sveriges Geologiska Undersökning Ca 53*.
Lundqvist, T., 1968: Precambrian geology of the Los-Hamra region, central Sweden. *Sveriges Geologiska Undersökning Ba 23*, 1–255.
Lundqvist, T. & Persson, P.-O., 1999: Geochronology of porphyries and related rocks in the northern and western Dalarna, south-central Sweden. *GFF 121*, 307–322.
Lundqvist, T., Gee, D.G., Kumpulainen, R., Karis, L. & Kresten, P., 1990: Beskrivning till berggrundskartan över Västernorrlands län. *Sveriges Geologiska Undersökning Ba 31*, 1–429.
Mattsson, H.J. & Elming, S.-Å., 2001: Magnetic fabrics and paleomagnetism of the Storsjön-Edsbyn deformation zone, central Sweden. *Precambrian Research 107*, 265–281.
McElhinny, M.W., 1973: *Paleomagnetism and plate tectonics*. Cambridge University Press. 358 pp.
Mellqvist, C., Öhlander, B., Skiöld, T. & Wikström, A., 1999: The Archaean-Proterozoic paleoboundary in the Luleå area, northern Sweden: field and isotope geochemical evidence for a sharp terrane boundary. *Precambrian Research 96*, 225–243.
Moakhar, M.O. & Elming, S.-Å., 1998: A paleomagnetic study of basic dykes and a gabbro formation in the Åsele and Hoting regions in central Sweden. *In* M.O. Moakhar: *Rapakivi granite and basic dykes in the Fennoscandian Shield: A paleomagnetic analysis*. Ph.D. thesis, Luleå University of Technology 1998:40.
Nyström, J.-O., 1982: Post-Svecokarelian Andinotype evolution in central Sweden. *Geologische Rundschau 71*, 141–157.
Nyström, J.-O., 1983: Pumpellyite-bearing rocks in Central Sweden and extent of host rock alteration as a control of pumpellyite composition. *Contributions to Mineralogy and Petrology 83*, 159–168
Patchett, P.J., Todt, W. & Gorbatschev, R., 1987: Origin of continental crust of 1.9–1.7 Ga Age: Nd isotopes in the Svecofennian orogenic terrains of Sweden. *Precambrian Research 35*, 145–160
Persson, P.-O. & Wikström, A., 1993: A U–Pb dating of the Askersund granite and its marginal augen gneiss. *Geologiska Föreningens i Stockholm Förhandlingar 115*, 321–329.
Pesonen, L.J. & Neuvonen, K.J., 1981: Paleomagnetism of the Baltic Shield – Implications for the Precambrian tectonics. *In* A. Kröner (ed.): *Pre-Cambrian Plate Tectonics*, 623-648. Elsevier.
Raposo, M.I.B., 1997: Magnetic fabric and its significance in the Florianópolis dyke swarm, southern Brazil. *Geophysical Journal International 131*, 159–170.
Skiöld, T., 1988: Implications of new U-Pb zircon chronology to early Proterozoic crustal accretion in northern Sweden. *Precambrian Research 38*, 147–164.
Suominen, V., 1991: The chronostratigraphy of southwestern Finland with special reference to Postjotnian and Subjotnian diabases. *Geological Survey of Finland Bulletin 356*, 1–100.
Tarling, D.H. & Hrouda F., 1993: *The magnetic anisotropy of rocks*. Chapman and Hall. 217 pp.
Tauxe, L., Gee, J.S. & Staudigel, H., 1998: Flow directions in dikes from anisotropy of magnetic susceptibility data: The bootstrap way. *Journal of Geophysical Research 103*, 17,775–17,790.
Welin, E. & Lundqvist, T., 1975: K-Ar ages of Jotnian dolerites in Västernorrland County, Central Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 97*, 83–88.
Welin, E., Christansson, K. & Kähr, A.-M., 1993: Isotopic investigations of metasedimentary and igneous rocks in the Paleoproterozoic Bothnian Basin, central Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 115*, 285–296.
Wikström, A., Sjöberg, B. & Kapicka, A., 1997: A precursory study of magnetic fabrics in the transition zone between magmatic and metamorphic textures in the Askersund granitoid, Sweden. *GFF 119*, 285–290.
Wilson, M.R., Hamilton, P.J., Fallick, A.E., Aftalion, M. & Michard, A., 1985: Granites and early Proterozoic crustal evolution in Sweden: evidence from Sm-Nd, U-Pb isotope systematics. *Earth and Planetary Science Letters 72*, 376–388.
Windley, B.F., 1993: Proterozoic anorogenic magmatism and its orogenic connections. *Journal of the Geological Society of London 150*, 39–50.